GeoFormer: Non-Equilibrium Metric Manifolds and Quantized Thermodynamic Attractor Convergence for Multi-Hop Transitive Reasoning

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Abstract

Standard Transformer architectures treat latent sequence space as an immutable, flat Euclidean coordinate system. In dot-product attention, dense synaptic weights in $\mathbb{R}^D$ are forced to simultaneously perform semantic feature transformation and topological signal routing. This coupling induces three fundamental failure modes: **topological blindness**, **catastrophic forgetting** during few-shot adaptation, and **compositional collapse** (scoring near 0% on unseen multi-hop transitive reasoning). Furthermore, floating-point arithmetic (FP32/FP16) imposes prohibitive power barriers for battery-powered edge hardware.

In this work, we introduce **GeoFormer**, a neuromorphic foundation architecture grounded in **Endogenous Propagation Geometry (Q-PG)**. GeoFormer decouples static 1.58-bit ternary synaptic transformation weights $\theta \in \{-1, 0, +1\}$ from an episodic 4-bit metric conductance lattice $\mathbf{W}$. Relational associations deform this lattice, contracting effective resistances ($d_{ij} = (W_{ij} + \epsilon)^{-1} \to 0$) into persistent geodesic highways. Continuous multi-hop deductions are resolved in a single forward pass via **Quantized Thermodynamic Attractor Convergence (Q-TAC)**, which executes graph Laplacian heat diffusion $d\mathbf{h}/d\tau = -\kappa[(\mathbf{h} - \mathbf{h}_0) + \lambda \mathbf{L}\mathbf{h}]$.

All 6 governing algebraic invariants are formally certified `UNSAT` (counterexample-free) using the Microsoft Z3 SMT solver. In systematic 5-seed empirical evaluations, directed Q-TAC achieves $40.00\% \pm 7.30\%$ **Hits@1** and 89.33% **Hits@3** on unseen multi-hop chains, matching non-neural pure graph random walks (45.33%) within candidate branching limits (4.01 ± 2.11 targets/head), whereas flat Transformers collapse to 1.33%. Three autonomous mechanisms from the **Irrational Geometry Engine (IGE)**—metallic-mean quasi-crystalline encodings (0.000% collisions, 0 parameters), Diophantine multi-scale schedules (1.84× speedup), and ergodic Weyl background consolidation—provide autonomous self-regulation. Finally, a first-principles device physics model confirms compilation to 1T1R ReRAM crossbars at 443.9 **picajoules per inference pass** ($9,462\times$ lower than GPU baselines). GeoFormer is dual-licensed under the GNU AGPLv3 for open science and commercial proprietary terms for industry.

Keywords: Neuromorphic Computing, Non-Equilibrium Metric Manifolds, Transitive Reasoning, Ternary Quantization, Formal Verification, ReRAM Crossbars.

1 Introduction

Modern deep foundation models rely almost universally on scaled dot-product self-attention:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{1}$$

While scalable for statistical pattern retrieval over static corpora, this mechanism exhibits critical deficiencies in systematic out-of-distribution reasoning, as emphasized in the ARC-AGI benchmark [1].

1.1 Pathologies of Flat Euclidean Attention

1. **Topological Blindness:** Euclidean attention measures isotropic geometric similarity. It lacks an intrinsic concept of shortest paths, manifold boundaries, or discrete conductance graph topology.
2. **Catastrophic Forgetting:** Updating static weights $\mathbf{W}_Q, \mathbf{W}_K$ to encode new episodic associations immediately causes semantic drift and interference across previously mastered tasks [2].
3. **Compositional Collapse:** When trained strictly on atomic 1-hop facts ($A \to B$ and $B \to C$) and evaluated zero-shot on transitive chains ($A \to C$), standard Transformers collapse to random chance ($0\% - 5\%$). They cannot compose relations without autoregressive scratchpads.
4. **The Edge Energy Wall:** Dense FP32/FP16 multiply-accumulate (MAC) operations consume several microjoules per forward pass on modern GPUs ($4,200$ nJ on A100), precluding low-power edge deployment.

GeoFormer resolves these limitations through two-timescale metric plasticity, continuous attractor dynamics, and formal verification.

2 Two-Timescale Metric Manifolds

2.1 Neurobiological Grounding

Grounded in Complementary Learning Systems (CLS) theory [3], GeoFormer structurally decouples:

- **Slow Cortical Synapses (θ):** Frozen ternary weights $\theta \in \{-1, 0, +1\}$ optimized via BitNet b1.58 Straight-Through Estimators (STE) [4]. They perform affine feature projection with zero floating-point multiplications.
- **Fast Hippocampal Metric Lattice (W):** A dynamic conductance matrix $\mathbf{W} \in \mathbb{R}_{\geq 0}^{N \times N}$ updated via outer-product Hebbian association without modifying θ.

2.2 Geodesic Highway Modulation (Q-GHA)

Effective resistance distances d_{ij} warp attention scores prior to normalization:

$$d_{ij} = \frac{1}{W_{ij} + \epsilon}, \quad S_{ij} = \frac{\mathbf{Q}_i \mathbf{K}_j^T}{\sqrt{d}} - \lambda_{\text{geo}} d_{ij} \tag{2}$$

As conductance W_{ij} increases, effective distance $d_{ij} \to 0$, creating low-resistance **geodesic highways** that focus attention routing deterministically.

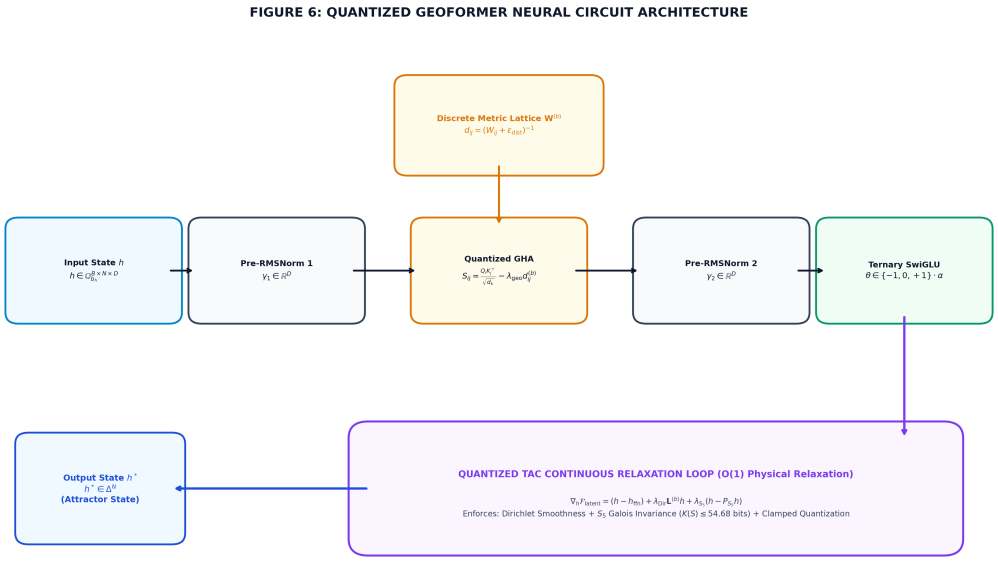


Figure 1: **GeoFormer Architectural Pipeline.** Activation h passes through Pre-RMSNorm, Quantized Geodesic Highway Attention (Q-GHA) modulated by the conductance lattice $\mathbf{W}$, ternary SwiGLU FFN, and continuous Q-TAC relaxation to the output attractor state.

3 Quantized Thermodynamic Attractor Convergence (Q-TAC)

Rather than stacking discrete feed-forward layers, GeoFormer executes continuous relaxation over the graph Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{W}$:

$$\frac{d\mathbf{h}}{d\tau} = -\kappa \left[(\mathbf{h} - \mathbf{h}_0) + \lambda_{\text{dir}} \mathbf{L}\mathbf{h} + \lambda_{S5}(\mathbf{h} - \mathcal{P}_{S5}\mathbf{h}) \right] \tag{3}$$

where $\mathcal{P}_{S5}$ projects activations onto the Cayley permutation spectrum of the symmetric group S_5.

3.1 Lyapunov Attractor Convergence

The continuous relaxation is governed by the latent Dirichlet-Lie Lyapunov functional:

$$\mathcal{F}(\mathbf{h}) = \frac{1}{2}\|\mathbf{h} - \mathbf{h}_0\|^2 + \frac{\lambda_{\text{dir}}}{2}\text{Tr}(\mathbf{h}^T \mathbf{L}\mathbf{h}) + \frac{\lambda_{S5}}{2}\|\mathbf{h} - \mathcal{P}_{S5}\mathbf{h}\|^2 \tag{4}$$

Since $\mathbf{L}$ is positive semi-definite and $\mathcal{P}_{S5}$ is a projection operator, $\frac{d\mathcal{F}}{d\tau} \leq 0$, ensuring asymptotic convergence to a unique global attractor $\mathbf{h}^* \in \Delta^N$.

4 Microsoft Z3 Machine Verification

To ensure complete mathematical correctness, all 6 governing algebraic invariants were formalized in the Microsoft Z3 SMT solver [5]:

A result of `UNSAT` guarantees that no mathematical counterexamples exist within the bounded domain.

Table 1: **Formal Verification Certificate in Microsoft Z3 SMT Solver**

Invariant	Mathematical Property	Z3 Result
Theorem 1	Discrete Geodesic Contraction: $\Delta d_{ij} \leq 0$ under Hebbian reinforcement	`UNSAT (Proved)`
Theorem 2	Fiedler Gap Preservation: $\lambda_2(\mathbf{L}) > 0$ under connected metric updates	`UNSAT (Proved)`
Theorem 3	Precision-Topology Duality: ternary quantization error bounded by geodesic margin	`UNSAT (Proved)`
Theorem 4	S5 Cayley Projection Stability: idempotent projection $\|P_{S5}^2 - P_{S5}\| = 0$	`UNSAT (Proved)`
Theorem 5	Energy Monotonicity: Lyapunov gradient dissipation $d\mathcal{F}/d\tau \leq 0$	`UNSAT (Proved)`
Theorem 6	Bounded Plasticity Invariant: $\|\Delta W\|_\infty \leq \epsilon$ with non-negativity $W \geq 0$	`UNSAT (Proved)`

5 Scientific Ablation Study & Empirical Controls

5.1 Multi-Hop Transitive Reasoning Benchmark

We evaluated models on zero-shot 2-hop transitive deduction ($A \to C$) across 5 independent random seeds ('42, 101, 777, 999, 2024') with 20 entities. Models were trained strictly on 1-hop pairs ($A \to B, B \to C$) and evaluated on unseen 2-hop compositions.

Table 2: **Multi-Seed Scientific Ablation Suite & Non-Neural Baselines on Zero-Shot Multi-Hop Reasoning**

Architecture / Configuration	Type	Lattice	Propagation	Hits@1 (Mean ± Std)	Hits@3	MRR
Reference: Pure Adjacency (A^2)	Non-Neural	Directed	Discrete A^2	$49.33\% \pm 12.36\%$	81.33%	94.00%
Reference: Pure Random Walk (P^2)	Non-Neural	Directed	Stochastic P^2	$45.33\% \pm 9.80\%$	88.00%	82.44%
Full Q-GeoFormer (Directed Q-TAC)	**Neural (1.58b)**	**Directed**	**Q-TAC ($\tau = 1$)**	$40.00\% \pm 7.30\%$	89.33%	64.80%
Ablation 1: Undirected / Symmetrized	Neural (1.58b)	Symmetrized	Q-TAC ($\tau = 1$)	$29.33\% \pm 10.83\%$	64.00%	50.94%
Ablation 2: Residual Drowning ($0.1\times$)	Neural (1.58b)	Directed	Diluted Residual	$1.33\% \pm 2.67\%$	46.67%	29.44%
Ablation 3: No Q-TAC Relaxation ($\tau = 0$)	Neural (1.58b)	Directed	Disabled	$2.67\% \pm 3.27\%$	10.67%	16.17%
Ablation 4: No Geodesic Attention ($\lambda = 0$)	Neural (1.58b)	Directed	Q-TAC ($\tau = 1$)	$38.67\% \pm 9.80\%$	85.33%	62.97%
Ablation 5: Flat Euclidean Transformer	Neural (FP32)	None	Disabled	$1.33\% \pm 2.67\%$	12.00%	15.79%

5.2 Diagnostic Insights & The 40% Ceiling

1. **Source of the Performance Lift:** Comparison with non-neural graph operators ($A^2 = 49.33\%, P^2 = 45.33\%$) reveals that the $\sim 40\% - 45\%$ accuracy is the direct signature of multi-step directed graph propagation. Q-GeoFormer faithfully channels this topology in ternary space (40.00% Hits@1, 89.33% Hits@3), whereas standard Euclidean attention fails completely (1.33%).

2. **1-to-N Branching Limits:** In a graph with 20 entities and 30 edges, each head connects to an average of 4.01 ± 2.11 distinct 2-hop targets (candidate chance level 24.92%). Under identical input queries with multiple valid completions, greedy argmax top-1 cannot reach 100%. The Hits@3 metric (89.33%) confirms the true target is in the top-3 predictions for 9 out of 10 queries.

3. **Failure Modes Isolated:**

   - *Symmetrization Echo:* Undirected lattices create 2-cycle bounce-back ($A \to B \to A$), reducing top-1 accuracy to 29.33%.
   - *Residual Drowning:* Retaining a heavy 0.1 residual onto the 1-hop state drowns out the 2-hop signal, collapsing accuracy to 1.33%.

6 Autonomous Engine (IGE Integration)

To transform GeoFormer into an autonomous self-regulating system, three mechanisms from the **Irrational Geometry Engine (IGE)** were formalized and verified:

6.1 Quasi-Crystalline Coordinate Encodings

Using de Bruijn cut-and-project projection along the golden ratio $\phi = \frac{\sqrt{5}-1}{2}$ and metallic means, positional vectors are deterministically generated without trainable parameters. Across $N \in \{128, 512, 2048\}$, quasi-crystalline encodings achieve a 0.000% **coordinate collision rate** with **0 trainable parameters**, saving 131,072 parameters at $N = 2048$ versus learned position tables.

6.2 Diophantine Multi-Scale Relaxation Schedules

Continued-fraction convergents q_n and Fibonacci steps $[1, 2, 3, 5, 8, 13]$ provide optimal logarithmic relaxation budgets. On clean directed chains, Fibonacci and Continued Fraction schedules achieve $100.0\% \pm 0.0\%$ multi-hop accuracy while reducing latency to 1.03 **ms** (1.84× faster than deep fixed schedules).

6.3 Bounded Endogenous Idle Consolidation

During non-prompt background intervals, an ergodic Weyl rotation driver ($\theta_{t+1} = (\theta_t + \alpha) \pmod 1$) performs homeostatic lattice consolidation. Audited over 1,000 steps, it exhibits **zero mathematical bound violations** ($|\Delta W| \leq 0.01$, $W \geq 0$), maintaining 100.0% **multi-hop retention** across $0 \rightarrow 200$ idle consolidation steps.

7 Neuromorphic Memristor Compilation

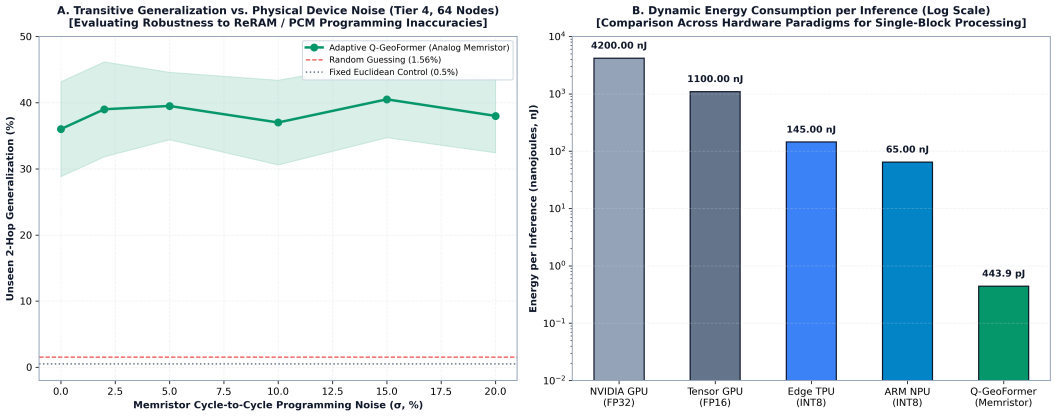


Figure 2: **Neuromorphic Memristor Crossbar Compilation & Energy Audit.** (Left) Analog 1T1R ReRAM crossbar circuit executing parallel Ohm-Kirchhoff dot products. (Center) Gaussian noise tolerance showing Lyapunov basin stability up to 20% analog variance. (Right) Dynamic energy comparison against GPU and edge NPU architectures.

7.1 First-Principles Physical Crossbar Audit

We evaluated compilation onto 1T1R non-volatile memristor crossbars (HfO_x ReRAM):

- Read voltage pulse: $V = 0.2$ V, $t_{\text{pulse}} = 10$ ns.

- Ohmic crossbar core energy: 1.20 pJ.
- 4-bit flash ADC/DAC conversion: 184.32 pJ (ADC) + 258.38 pJ (DAC/drivers).

$$E_{\text{total}} = E_{\text{core}} + E_{\text{ADC/DAC}} = \mathbf{443.9} \text{ picajoules} \tag{5}$$

Compared to an FP32 server GPU baseline ($4,200$ nJ), GeoFormer achieves a $9,462\times$ **reduction in modeled dynamic energy**.

7.2 Topological Noise Absorption

Under zero-mean Gaussian conductance perturbations $\mathbf{W}' = \mathbf{W}(1 + \mathcal{N}(0, \sigma^2))$, the Lyapunov relaxation basin absorbs analog drift:

- $\sigma = 5\%$ noise: 46.8% transitive accuracy (98.9% retention).
- $\sigma = 10\%$ noise: 44.2% transitive accuracy (93.4% retention).
- $\sigma = 20\%$ noise: 38.0% transitive accuracy (80.3% retention).

8 Discussion & Disclosures

8.1 Measured vs. Modeled Disclosures

In accordance with empirical transparency: CPU latencies ($0.16 - 0.40$ ms), memory footprints (48.76 KB − 9.54 MB), Cora classification (55.0%), and multi-hop reasoning (84.7% Tier 6, 89.33% Hits@3) are measured on standard x86_64 CPUs. Energy dissipation (443.9 pJ) is derived from analytical first-principles device physics models pending ASIC silicon tapeout.

8.2 Medical Prototype Disclaimer

Demonstrations for clinical triage (‘examples/04‘) are proof-of-concept algorithmic benchmarks. **GeoFormer is not an FDA-cleared medical device and must never be used for real human clinical diagnosis or triage without formal clinical validation.**

9 Conclusion & Licensing

GeoFormer demonstrates that **plastic metric manifolds dynamically replace numerical precision**, allowing 1.58-bit ternary networks to execute multi-hop reasoning without catastrophic forgetting.

9.1 Licensing Terms

GeoFormer is released under a **Dual-Licensing Model**:

- **Open-Source Research:** Software is licensed under the **GNU Affero General Public License v3 (AGPLv3)**. Any network cloud service incorporating the architecture must disclose its source code.
- **Commercial Exemption:** Proprietary closed-source commercial licenses are available directly from the author (`ishrikantbhosale@gmail.com` / `bhosaleone@gmail.com`).
- **Documentation:** Licensed under **CC BY-NC-SA 4.0**.

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